

KAREEM EL-BADRY

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RESEARCH INTERESTS

binary stars, stellar-mass black holes, neutron stars, white dwarfs, wide-field surveys;
near-field cosmology, galactic archaeology, globular clusters;
galaxy formation, low-mass galaxies, stellar feedback

EDUCATION

Ph.D., Astrophysics, University of California, Berkeley	2021
Thesis: <i>Binary stars across the Milky Way: Probes of star formation and evolution</i>	
M.A., Astrophysics, University of California, Berkeley	2018
B.S., Astronomy & Physics, <i>summa cum laude</i> , Yale University	2016

RESEARCH POSITIONS

Associate Professor, California Institute of Technology	2026–
Adjunct Scientist, MPIA, Heidelberg	2023–
Assistant Professor, California Institute of Technology	2023 – 2026
Junior Fellow, Harvard Society of Fellows	2021 – 2023
ITC Fellow, Harvard-Smithsonian Center for Astrophysics	2021 – 2023
Postdoctoral Fellow, recurring summer appointment, MPIA, Heidelberg	2021 – 2023
Graduate Student, University of California, Berkeley	2016 – 2021
Kavli Summer Research Fellow, CCA, NYC	2018
Summer Visiting Researcher, MPIA, Heidelberg	2017 – 2020
Summer Undergraduate Research Fellow, Caltech	2015
Undergraduate Research Assistant, Yale	2015 – 2016
Dean's Summer Research Fellow, Yale	2014

HONORS & AWARDS

Eddington Lectureship, University of Cambridge & Royal Astronomical Society	2026
MacArthur Fellowship	2025
Alfred P. Sloan Research Fellowship	2025
New Horizons Solvay Lecturer in Physics	2025
IAU Ph.D. Prize, Stars and Stellar Physics division	2022
Mary Elizabeth Uhl Dissertation Prize, Berkeley	2021
Outstanding Graduate Student Instructor Award, Berkeley	2021
Robert J. Trumpler Graduate Student Excellence Award, Berkeley	2020
CCAPP Prize Prize in Cosmology and AstroParticle Physics	2018
NSF Graduate Research Fellowship	2016 – 2021
Berkeley Fellowship	2016 – 2018
Hellman Award for Graduate Study	2016 – 2018
George Beckwith Prize in Astronomy, Yale	2016
Phi Beta Kappa, Yale	2015
Jerry Inskeep Memorial Scholarship, Yale	2014

REFEREED PUBLICATIONS (SUMMARY)

201 total; 41 first-author
h-index: 54 (all papers), 32 (first-author papers)
citations: 9300+ (all papers), 3600+ (first-author papers)

GRANTS

Roman General Investigator Grant , NASA/STScI, Co-PI, Total: \$450,000 Roman-19028: <i>RACS: Roman Astrometric Companion Survey</i> Co-PI: Cheyanne Shariat	2027-2030
Roman General Investigator Grant , NASA/STScI, Co-PI, Total: \$450,000 Roman-19046: <i>Binary Star Demographics in the Galactic Bulge</i> Co-PI: Cheyanne Shariat	2027-2030
Roman General Investigator Grant , NASA/STScI, Co-PI, Total: \$450,000 Roman-19080: <i>RAGHAB: Roman Analysis for a Galactic Hunt for Astrometric Binaries</i> Co-PI: Soumyadeep Bhattacharjee	2027-2030
JWST General Observer Grant , NASA/STScI, PI, Total: \$197,655 JWST-GO-10493: <i>Weighing the heaviest neutron stars with JWST</i>	2026-2029
JWST General Observer Grant , NASA/STScI, PI, Total: \$205,225 JWST-GO-11051: <i>Imaging and spectroscopy of a hypervelocity globular cluster ejected from M87</i> Science PI: Cheyanne Shariat	2026-2029
JWST General Observer Grant , NASA/STScI, CO-I, subaward: \$17,500 JWST-GO-12507: <i>A Panoramic View of Rapid Dust Production by Disintegrating Asteroids around a White Dwarf</i>	2026-2027
CAREER Grant , NSF, PI, Total: \$886,226 NSF-AST-2540180: <i>CAREER: Mapping binary evolution across the Milky Way with Gaia</i>	2026-2031
HST General Observer Grant , NASA/STScI, PI, Total: \$66,324 HST-GO-18120: <i>A COS Legacy Survey of Gaia DR3 WD+MS binaries</i> Science PI: Natsuko Yamaguchi	2026-2029
Astronomy and Astrophysics Research Grant , NSF, PI, Total: \$496,984 NSF-AST-2508988: <i>Hypervelocity survivors of thermonuclear supernovae: new models and new discoveries</i>	2025-2028
Sloan Fellowship , physics, Alfred P. Sloan Foundation, PI, \$75,000	2025-2027
Scialog Grant , RCSA, CO-PI, \$66,000 to Caltech SA-LSST-2024-114c: <i>IMBH in the LMC? A hypervelocity star survey with LSST</i>	2025-2027
Scialog Grant , RCSA and Heising-Simons Foundation, CO-PI, \$66,000 to Caltech SA-LSST-2024-084a: <i>White dwarf companions as brown dwarf chronometers</i>	2025-2027
JWST General Observer Grant , NASA/STScI, PI, Total: \$79,486 JWST-GO-04979: <i>Uncovering the cold donors of AM CVn binaries</i>	2025-2027
HST General Observer Grant , NASA/STScI, PI, Total: \$88,306 HST-GO-17441: <i>UV spectroscopy of runaways from thermonuclear supernovae</i>	2024-2026
Astronomy and Astrophysics Research Grant , NSF, PI, Total: \$475,345 NSF-AST-2307232: <i>Dormant black holes and neutron stars in stellar binaries</i>	2023-2026
HST General Observer Grant , NASA/STScI, PI, Total: \$30,000 HST-GO-17202: <i>Does MWC 656 host a black hole or a stripped helium star?</i>	2023-2025
HST General Observer Grant , NASA/STScI, PI, Total: \$48,171 HST-GO-17200: <i>A census of massive binaries and LBVs in the Whirlpool Galaxy</i>	2023-2025
HST General Observer Grant , NASA/STScI, CO-I, subaward: \$23,181 HST-GO-17216: <i>Establishing a 1% anchor for the Population II Distance Ladder with HST</i>	2023-2025

AWARDED TELESCOPE TIME (SUMMARY)

HST	19 orbits (PI) + 148 orbits (CO-I)
JWST	23 hours (PI) + 82 hours (CO-I)
Keck 10m	23 nights (PI) + 26 nights (CO-I)
Magellan 6.5m	9 nights (PI) + 4 nights (CO-I)
VLT 8.2m	46 hours (PI) + 40 hours (CO-I)
LBT 2 × 8.4m	3 nights (PI) + 3 nights (CO-I)
Gemini 8.1m	23 hours (PI)
Palomar Hale 200 inch	5 nights (PI) + 29 nights (CO-I)
MPG/ESO La Silla 2.2m	55 nights (PI) + 50 nights (CO-I)
Tillinghast 1.5m FLWO	30 nights (PI) + 10 nights (CO-I)
Las Cumbres Observatory 2×1 m	1.5 nights (PI)
Lick Shane 3m	26 nights (PI)
Lick APF 2.4m	15 nights (CO-I)
Nordic Optical Telescope 2.6m	12 nights (CO-I)
NTT 3.6m	3 nights (CO-I)
McDonald 2.7m Harlan J. Smith Telescope	5 nights (CO-I)
Very Large Array	4 hours (DDT; CO-I)
Green Bank Telescope	12 hours (CO-I)
MeerKAT	4 hours (DDT; CO-I)
Chandra	20ks (DDT; PI) + 20ks (DDT; CO-I)

COLLABORATION MEMBERSHIPS

<i>Euclid</i> Consortium (MW Spectroscopy working group)	2025 –
<i>Gaia</i> Data Processing and Analysis Consortium (CU4: non-single stars)	2024 –
<i>UVEX</i> mission (Low-mass, low-metallicity galaxies & the Magellanic Clouds)	2021 –
Sloan Digital Sky Survey (binaries & compact objects working groups)	2021 –

CONFERENCE ORGANIZING

Lead organizer of StellarBH25 (8-week KITP meeting on stellar-mass black holes), Santa Barbara	2025
Co-chair of Keck science meeting, UCLA	2025
SOC of Binary stars in the space era conference, Keele University	2025

JOURNAL REFEREE

A&A, A&AL, ApJ, ApJL, MNRAS, MNRASL, SCPMA, Nature, Nature Astronomy, OJAp	42 papers; 2017 –
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MENTORING

El-Badry group at Caltech

Leo Betancourt-Chitnis (PhD student)	2025–
Soumyadeep Bhattacharjee (PhD student)	2024–
Cheyenne Shariat (PhD student)	2024–
Lisa Blomberg (PhD student)	2023–
Natsuko Yamaguchi (PhD student)	2022–
Pranav Nagarajan (PhD student)	2022–
Maude Gull (Brinson Postdoctoral Fellow)	2025–
Geoffrey Mo (Brinson Postdoctoral Fellow)	2025–
Joe Smith (Undergrad)	2025–
Avni Bansal (Undergrad)	2025–
Antonio Rodriguez (PhD student)	2022 – 2025
Domani Sharkey (SURF student)	2024 – 2025
Evan Portnoi (SURF student)	2023 – 2024
Umran Koca (Undergrad)	2023 – 2024

Elsewhere

Ben Pennell (MPIA PhD student)	2024–
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Johanna Müller-Horn (MPIA PhD student)	2023–
Casey Lam (Carnegie Observatories postdoc)	2023–
Rhys Seeburger (MPIA PhD student)	2021 – 2025
Maude Gull (Berkeley PhD student)	2022 – 2025
Jesse Han (Harvard PhD student)	2022 – 2023
Vedant Chandra (Harvard PhD student)	2021 – 2023
Silvia Almada (MPIA masters student)	2020 – 2022
Pranav Nagaraajan (Berkeley undergrad)	2020 – 2022
Nick Choksi (Berkeley undergrad)	2018 – 2019

TEACHING EXPERIENCE

Instructor, Ay 20, <i>Basic astronomy and the Galaxy</i> , Caltech	FA2026
Instructor, Ay 125, <i>High energy astrophysics</i> , Caltech	WI2026
Instructor, Ay 124, <i>Structure and Evolution of Galaxies</i> , Caltech	WI2025, WI2027
Instructor, Ay 03, <i>Automating Discovery in the Universe</i> , Caltech	SP2024
Instructor, Ay 123, <i>Stellar Structure and Evolution</i> , Caltech	WI2024, FA2024
Graduate Student Instructor, Astro 128, <i>Astronomy Data Lab</i> , University of California, Berkeley	2019, 2020, 2021
Course Designer, Astro 128, <i>Astronomy Data Lab</i> , University of California, Berkeley	2018, 2019, 2020
Co-Instructor, Astro 375, <i>Graduate Pedagogy</i> , University of California, Berkeley	2019, 2020
Sole Instructor, <i>Stellar Physics</i> , Hyeonpung High School, Daegu, South Korea	2019
Graduate Student Instructor, Astro 160, <i>Stellar Physics</i> , University of California, Berkeley	2018
Graduate Student Instructor, Astro 7A, <i>Introduction to Astronomy</i> , University of California, Berkeley	2017
Graduate Student Instructor, Astro C12, <i>The Planets</i> , University of California, Berkeley	2017
Tutor & Grader, Math 120, <i>Multivariable Calculus</i> , Yale	2013 – 2016
Tutor, Math 111, <i>College Algebra</i> , Umpqua Community College	2013

TALKS (SINCE 2021)

Talk, EUROWD26, Vienna, Austria	2026
Talk, Unsolved Problems in Astrophysics and Cosmology workshop, Vienna, Austria	2026
Heidelberg joint astronomy colloquium, Heidelberg, Germany	2026
Physics colloquium, University of California, Davis	2026
Physics colloquium, University of Southern California, Los Angeles, CA	2026
Eddington Lecture, Royal Astronomy Society, London, UK	2026
Eddington Lecture, Institute of Astronomy, Cambridge, UK	2026
Invited talk, MM2026 Yukawa International Seminar, YITP, Kyoto, Japan	2026
Review talk, StellarBH25, Kavli Institute for Theoretical Physics, Santa Barbara, CA	2025
Astronomy colloquium, Johns Hopkins University and STScI, Baltimore, MD	2025
Königstuhl colloquium, Max Planck Institute for Astronomy, Heidelberg, Germany	2025
Talk, Binary Stars in the Space Era conference, Keele, UK	2025
Astrophysics seminar, Royal Observatory of Belgium, Brussels, Belgium	2025
Astrophysics seminar, University of Liège, Liège, Belgium	2025
New Horizons Solvay Lectures in Physics, Université libre de Bruxelles, Brussels, Belgium	2025
KIPAC astrophysics colloquium, Stanford University, Palo Alto, CA	2025
CfA colloquium, Center for Astrophysics Harvard & Smithsonian, Cambridge, MA	2025
Astronomy colloquium, Columbia University, New York City, NY	2025
Talk, 10 years to LISA meeting, Jet Propulsion Laboratory, Pasadena, CA	2025
Astrophysics colloquium, Princeton University, Princeton, NJ	2025
Astrophysics seminar, Carnegie Mellon University, Pittsburgh, PA	2025
Astronomy colloquium, Indiana University, Bloomington, IN	2025
Astronomy colloquium, University of Texas, Austin, TX	2024
Review talk, Multimessenger binary mergers meeting, Aspen, CO	2024
Review talk, Lorentz center workshop on stripped stars, Leiden, Netherlands	2024
Talk, EUROWD24, Barcelona, Spain	2024
Talk, SDSS-V collaboration meeting, Las Cruces, NM	2024
Astronomy colloquium, University of Washington, Seattle, WA	2024
Talk, 20 Years of the ITC conference, Cambridge, MA	2024
IAS Astrophysics Seminar, Princeton, NJ	2024

Caltech Astronomy Colloquium, Pasadena, CA	2024
UCSC Astronomy Colloquium, Santa Cruz, CA	2023
Sydney Institute for Astronomy seminar, Sydney, AU	2023
Astrophysics colloquium, Carnegie Observatories, Pasadena, CA	2023
Review talk, Massive Triples, Binaries and Mergers, Leuven, Belgium	2023
Review talk, Lorentz center workshop on stellar black holes, Leiden, Netherlands	2023
Astrophysical sciences division colloquium, NASA/Goddard	2023
Astrophysics colloquium, Boston University	2023
Astrophysics colloquium, Cornell University	2023
Astrophysics colloquium, University of Pennsylvania	2023
Astrophysics colloquium, University of Michigan	2022
Talk, KITP white dwarf conference, Santa Barbara, CA	2022
Astrophysics colloquium, Dunlap Institute, University of Toronto	2022
Astrophysics colloquium, Yale University	2022
Talk, ZTF theory network meeting, CA	2022
Talk, EUROWD22, Tübingen, Germany	2022
Galaxy coffee, Max Planck Institute for Astronomy, Heidelberg, Germany	2022
Astrophysics colloquium, Shanghai Astronomical Observatory	2022
Astrophysics colloquium, University of California, Los Angeles	2022
IAU working group on active B stars seminar	2022
Albert Einstein Institute seminar, Potsdam, Germany	2022
UCSD/SDSU joint colloquium, San Diego	2021
Harvard Society of Fellows lunch talk, Harvard University	2021
Stellar Astrophysics Center seminar, Aarhus University, Aarhus, Denmark	2021
Astrophysics colloquium, Massachusetts Institute of Technology	2021
Institute of Theory and Computation colloquium, Harvard University	2021
Königstuhl colloquium, Max Planck Institute for Astronomy, Heidelberg, Germany	2021
Astrophysics colloquium, CIERA center, Northwestern University	2021
Astronomy seminar, University of Warwick, UK	2021
Lunch talk, University of California, Berkeley	2021
Astronomy colloquium, University of California, Berkeley	2021
Astronomy colloquium, California Institute of Technology	2021
Astronomy colloquium, University of Chicago	2021
Physics colloquium, University of Utah	2021